

## Methods

Baseline characteristics of study participants were summarized by diabetes status. Continuous variables are presented as mean (standard deviation) and categorical variables are presented as frequency and percentage. Missing values are reported where applicable. Kaplan-Meier survival curves were estimated separately for participants with and without diabetes to examine time to first myocardial infarction or fatal coronary heart disease (MI/CHD).

### Statistical Analysis

Median survival time and survival probabilities at 5, 10, 15, and 20 years were estimated with 95% confidence intervals for each group and compared using the log-rank test. To account for multiple comparisons, a Bonferroni-corrected significance level of  $\alpha^* = 0.01$  was applied. Univariate Cox proportional hazards models were used to assess the associations of diabetes and total cholesterol with MI/CHD risk. Total cholesterol was centered at the sample mean and expressed per 10 mg/dL increase. Martingale residuals with LOWESS smoothing were used to evaluate the linearity assumption for total cholesterol, and a log transformation was applied to achieve linearity. A multivariable Cox model was then developed including both exposures and potential confounders (age, sex, education), retained if they produced  $\geq 10\%$  change in the hazard ratio. The proportional hazards assumption was assessed using Schoenfeld residuals. Model diagnostics included Cox-Snell, deviance residuals, and DFBETA statistics. Effect modification by diabetes status on the cholesterol-MI/CHD association was assessed via an interaction term. In a secondary analysis, cholesterol was categorized as healthy ( $<200$  mg/dL), at risk ( $200\text{--}239$  mg/dL), or dangerous ( $\geq 240$  mg/dL), with unadjusted and adjusted Cox models estimated. Kaplan-Meier curves and a trend test were used to assess dose-response. All analyses were conducted using Stata version 18.5. A two-sided  $\alpha = 0.05$  was used unless otherwise specified.

## Results

Baseline characteristics of the 4,354 study participants by diabetes status are presented in Table 1. The majority of participants did not have diabetes ( $n = 4,240$ , 97.4%), with 114 participants (2.6%) having diabetes at baseline. On average, diabetic participants were older (55.2 vs. 49.7 years) and had higher total cholesterol levels (248.1 vs. 236.6 mg/dL) compared to non-diabetic participants. Among diabetic participants, a higher proportion were male (50.0% vs. 42.9%) and had some high school or less education (55.3% vs. 40.6%).

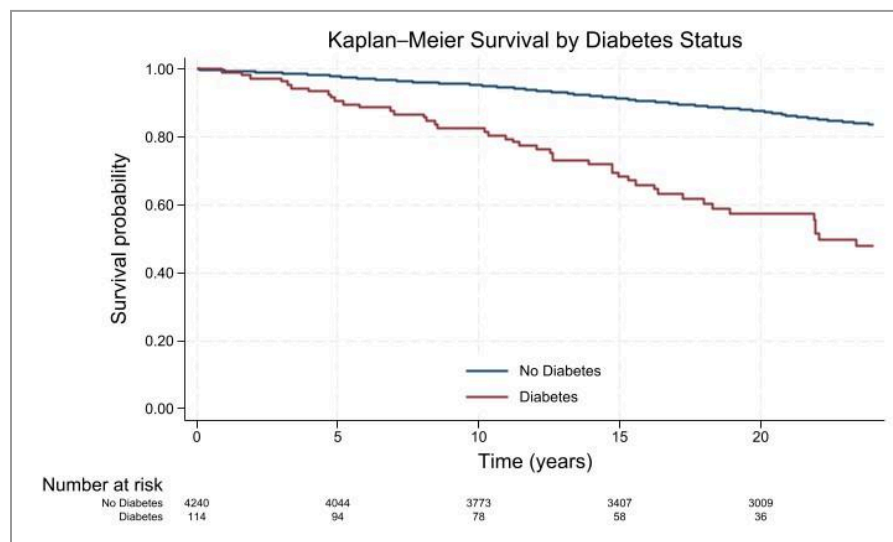
**Table 1.** Baseline characteristics of Framingham Heart Study participants by diabetes status, 1948 (N = 4,354).

| Characteristic                         | No Diabetes (N=4240) | Diabetes (N=114) | Total        |
|----------------------------------------|----------------------|------------------|--------------|
| Age (years)                            | 49.7 (8.6)           | 55.2 (7.6)       | 49.8 (8.6)   |
| Total cholesterol (mg/dL) <sup>a</sup> | 236.6 (44.3)         | 248.1 (56.3)     | 236.9 (44.6) |
| Sex                                    |                      |                  |              |
| Male                                   | 1821 (42.9%)         | 57 (50.0%)       | 1878 (43.1%) |
| Female                                 | 2419 (57.1%)         | 57 (50.0%)       | 2476 (56.9%) |
| Education, n (%) <sup>b</sup>          |                      |                  |              |
| Some HS or less                        | 1722 (40.6%)         | 63 (55.3%)       | 1785 (41.0%) |
| High school/GED                        | 1244 (29.3%)         | 24 (21.1%)       | 1268 (29.1%) |
| Some college                           | 689 (16.2%)          | 15 (13.2%)       | 704 (16.2%)  |
| College graduate+                      | 477 (11.2%)          | 9 (7.9%)         | 486 (11.2%)  |

Values are presented as mean (SD) for continuous variables and n (%) for categorical variables.

<sup>a</sup> Total cholesterol missing for 52 participants. <sup>b</sup> Education missing for 111 participants.

The Kaplan-Meier survival curves by diabetes status is shown in Figure 1. The median survival time at 50% was not reached for those without diabetes at baseline. However, 50% of diabetic participants experienced their first myocardial infarction event or coronary heart disease death by approximately 22.1 years (95% CI: 17.97, not estimable, Figure 1), as the upper confidence limit could not be estimated within the observed follow-up period and reflecting a substantially higher event burden in this group.

**Figure 1.** Kaplan-Meier survival curves for time to first myocardial infarction or coronary heart disease death, by diabetes status.

The number at risk at 0, 5, 10, 15, and 20 years is shown below in the figure.

**Table 2.** Survival estimates and time-point tests.

| Time  | No Diabetes S(t) (95% CI) | Diabetes S(t) (95% CI) | $\chi^2(1)$ | p-value | Sig.? |
|-------|---------------------------|------------------------|-------------|---------|-------|
| 5 yr  | 0.97 (0.96, 0.98)         | 0.90 (0.83, 0.94)      | 98.06       | <0.001  | Yes   |
| 10 yr | 0.95 (0.94, 0.95)         | 0.82 (0.73, 0.88)      | 100.64      | <0.001  | Yes   |
| 15 yr | 0.91 (0.90, 0.92)         | 0.68 (0.57, 0.76)      | 103.98      | <0.001  | Yes   |
| 20 yr | 0.87 (0.86, 0.88)         | 0.57 (0.46, 0.67)      | 123.37      | <0.001  | Yes   |

*Estimated survival probabilities with 95% confidence intervals at 5, 10, 15, and 20 years of follow-up, by diabetes status. Log-rank tests at each time point, assessed using a Bonferroni-corrected significance threshold of  $\alpha^* = 0.01$ . S(t) = survival probability at time t; CI = confidence interval.*

Estimated survival probabilities at 5, 10, 15, and 20 years are presented in Table 2. Survival was consistently lower among diabetic participants at all time points, ranging from 0.90 (95% CI: 0.83, 0.94) at 5 years to 0.57 (95% CI: 0.46, 0.67) at 20 years. In contrast, non-diabetic participants maintained markedly higher survival throughout follow-up, from 0.97 (95% CI: 0.97, 0.98) at 5 years to 0.87 (95% CI: 0.86, 0.88) at 20 years. Notably, the absolute difference in survival between groups widened over time, approximately 7 percentage points at 5 years and 31 percentage points at 20 years, suggesting that the survival disadvantage associated with diabetes accumulates over longer follow-up. Log-rank tests conducted at each time point confirmed that differences in survival between diabetic and non-diabetic participants were statistically significant at 5, 10, 15, and 20 years (all  $\chi^2(1) > 97$ ,  $p < 0.001$ ), and remained so after Bonferroni correction for multiple comparisons with  $p^* = 0.01$ .

The overall log-rank test further confirmed a statistically significant difference in survival across the full follow-up period ( $\chi^2(1) = 97.29$ ,  $p < 0.001$ ), with diabetic participants experiencing a disproportionately higher number of observed events relative to what would be expected under the null hypothesis of equal survival (44 observed vs. 11.24 expected), while non-diabetic participants experienced fewer events than expected (607 observed vs. 639.76 expected).

Univariate analysis of diabetes indicated that there is a significant association between being diabetic and hazard of MI/CHD. Compared to non-diabetics, participants with diabetes had approximately 4 times the hazard of MI/CHD (HR = 4.14; 95% CI: 3.05 - 5.63;  $p < 0.001$ ). A significant association was also observed between total cholesterol and MI/CHD. Specifically, after centering around the sample mean to account for a wide range of values, a 10 mg/dL increase in total cholesterol levels was significantly associated with a 7% increase in hazard of MI/CHD (HR = 1.07; 95% CI: 1.06 - 1.09;  $p < 0.001$ ).

Primary assessment of linearity using martingale residuals plot of centered total cholesterol levels showed the LOWESS line curving upward as cholesterol increases, and then an abrupt change, suggesting that there is non-linearity in the relationship between cholesterol and the log hazard (Figure S1, Panel A). To rectify this non-linearity, a subsequent attempt using log-transformed total cholesterol was performed and showed a LOWESS line with the slope

changing across the range, although not as drastically as in the first attempt (Figure S1, Panel B). Given that the curvature was mild and the abrupt change in slope is gone, it was concluded that log transformation improved linearity and was retained in the model. Accordingly, log-transformed total cholesterol was retained in all subsequent Cox models.

Hazard ratios for the base model and models with potential confounding variables are depicted in Table S1. Age and sex were identified as confounders, as their inclusion resulted in a meaningful change (>10%) in the hazard ratio for diabetes and log-transformed total cholesterol. Education did not substantially change the estimates and therefore were not retained in the final model.

**Table 3.** Univariate and multivariable Cox proportional hazards models for risk of MI/CHD

| Model                                    | Univariate Model | p-value | Final Multivariable Model <sup>a</sup> | p-value |
|------------------------------------------|------------------|---------|----------------------------------------|---------|
| <b>Diabetes (Yes vs. No)</b>             | 4.14 (3.04–5.63) | <0.001  | 3.05 (2.23–4.10)                       | <0.001  |
| <b>Total cholesterol (mg/dL)</b>         |                  |         |                                        |         |
| Main Effect                              | 1.07 (1.05–1.08) | <0.001  | 18.73 (6.37–55.12)                     | <0.001  |
| TVC                                      | —                | —       | 0.64 (0.41–0.99)                       | <0.001  |
| <b>Age (per year)</b>                    | —                | —       | 1.05 (1.04–1.06)                       | <0.001  |
| <b>Sex (Female vs. Male)<sup>a</sup></b> | —                | —       | —                                      | —       |

HR (CI); HR = hazard ratio.; TVC = Time-varying covariate

<sup>a</sup> Stratified by sex, therefore no HR provided for sex.

An intermediate, multivariable Cox model included diabetes, log-transformed total cholesterol, age, and sex. In this model, diabetes was associated with a 3.06-fold increase in the hazard of MI/CHD (HR = 3.05, 95% CI: 2.23–4.18,  $p < 0.001$ ). A 1-unit increase in log total cholesterol was associated with a significantly higher hazard of MI/CHD (HR = 6.63; 95% CI: 4.29 – 10.23;  $p < 0.001$ ). Each additional year of age was associated with a 5% increase in MI/CHD hazard (HR = 1.05; 95% CI: 1.04 – 1.06;  $p < 0.001$ ). Compared to males, females had significantly lower MI/CHD hazard (HR = 0.31; 95% CI: 0.26 – 0.36;  $p < 0.001$ ).

The intermediate, multivariable Cox proportional hazards model is as follows:

$$h(t) = h_0(t) \exp(\beta_1 \cdot I(\text{diabetes} = \text{Yes}) + \beta_2 \cdot \log(\text{total cholesterol}) + \beta_3 \cdot \text{age} + \beta_4 \cdot I(\text{sex} = \text{Women}))$$

The proportional hazards assumption was evaluated for the intermediate model using Schoenfeld residuals. Evidence of assumption violation was observed for sex ( $p = 0.151$ ,  $\chi^2(1) =$

13.60,  $p < 0.001$ ), and marginally for log total cholesterol ( $p = -0.08$ ,  $\chi^2(1) = 3.81$ ,  $p = 0.051$ ). Although no evidence of violation was observed for diabetes ( $p = 0.04$ ,  $\chi^2(1) = 0.90$ ,  $p = 0.34$ ) and age ( $p = -0.07$ ,  $\chi^2(1) = 3.20$ ,  $p = 0.07$ ), the global test was significant ( $\chi^2(4) = 19.13$ ,  $p < 0.001$ ), providing sufficient evidence that the proportional hazards assumption was violated across the model as a whole.

After stratification by sex, the global test of Schoenfeld residuals still indicated evidence of violation of the proportional hazards assumption ( $\chi^2(3) = 8.53$ ,  $p = 0.04$ ). To address this, log-transformed total cholesterol was included as a time-varying covariate in the Cox model. The interaction between log-transformed cholesterol and log(time) was statistically significant ( $p = 0.044$ ), indicating that the effect of total cholesterol on MI/CHD hazard varies over time.

The final multivariable Cox model included diabetes, age, and time-varying total cholesterol, with stratification by sex. In this model, diabetes was associated with a 3-fold increase in the hazard of MI/CHD (HR = 3.0, 95% CI: 2.19–4.10,  $p < 0.001$ ). The hazard ratio for total log-transformed cholesterol at baseline was 18.73 (95% CI: 6.37–55.12,  $p < 0.001$ ), however this effect was found to decrease with each additional year of follow-up (tvc HR = 0.64, 95% CI: 0.41–0.99,  $p < 0.001$ ). A one-year increase in age was significantly associated with a 5% increased hazard (HR = 1.05, 95% CI: 1.04–1.06,  $p < 0.001$ ). Estimates and 95% confidence intervals from univariate and multivariable Cox proportional models are presented in Table 3.

The final multivariable, stratified extended Cox model is as follows:

$$h_{\text{sex}}(t | X) = h_{0,\text{sex}}(t) \exp [\beta_1 I(\text{diabetes} = \text{Yes}) + \beta_2 \text{age} + \beta_3 \log(\text{total cholesterol}) + \beta_4 \log(\text{total cholesterol}) \cdot \log(t)]$$

Martingale residual plots for age showed no evidence of non-linearity, as the LOWESS-smoothed curve was approximately linear (Figure S2, Panel A). Therefore, age was retained as a linear term in the model. Cox-Snell residual plots indicated that overall model fit improved with the inclusion of additional covariates (Figure S2, Panel B). Simpler models, particularly those including only a single predictor, showed noticeable deviations from the 45-degree reference line, suggesting poorer fit. Adding both diabetes and log-transformed cholesterol improved alignment with the reference line, and the inclusion of age further improved the fit across most of the residual range. The best fit was observed for the model stratified by sex, which showed the closest agreement with the 45-degree line over the majority of the distribution. Although some deviation was still present at the right tail across all models, this is expected and likely reflects sparse data and instability in the tail rather than substantial model misspecification.

Deviance residual plots showed that most observations were centered around zero with no strong systematic pattern over time, suggesting reasonable model fit (Figure S2, Panel C). A small number of observations exhibited larger residual values, indicating potential outliers, and a slight downward trend was observed over time. However, no substantial departures from model assumptions were evident.

DFBETA plots indicated that individual observations had minimal influence on model coefficients (Figure S2, Panel D). Values were centered around zero with no systematic pattern over time. A small number of observations showed a slightly larger influence for log-transformed cholesterol, but no extreme values were observed. Overall, the model appeared stable and not unduly influenced by individual subjects.

A secondary analysis was conducted to assess whether the association between total cholesterol and MI/CHD hazard varied by diabetes status. The interaction term between diabetes and log-transformed total cholesterol was not statistically significant (HR = 3.71; 95% CI: 0.66–20.84;  $p = 0.136$ ), indicating no evidence that the effect of log-transformed total cholesterol on MI/CHD hazard differs significantly by diabetes status. This suggests that the association between total cholesterol and MI/CHD hazard is not significantly different among individuals with and without diabetes.

In an additional analysis, total cholesterol was categorized into three clinically relevant groups: <200 mg/dL (healthy), 200–239 mg/dL (at risk), and  $\geq 240$  mg/dL (dangerous). In the unadjusted analysis, compared to individuals with healthy cholesterol levels, those with at-risk and dangerous levels had significantly higher hazards of MI/CHD (HR = 1.75, 95% CI: 1.33–2.30; and HR = 2.54, 95% CI: 1.96–3.28, respectively; Table S2). After adjustment for diabetes status and age, and stratifying by sex, the associations were attenuated but remained statistically significant. Compared to the healthy reference group, individuals with cholesterol levels of 200–239 mg/dL had a hazard ratio of 1.53 (95% CI: 1.17–2.01), and those with levels  $\geq 240$  mg/dL had a hazard ratio of 2.22 (95% CI: 1.71–2.87; Table S2). Survival curves differed significantly across the three cholesterol categories (Figure S3; log-rank  $\chi^2(2) = 60.57$ ,  $p < 0.001$ ), with a significant increasing trend in MI/CHD hazard with higher cholesterol category (log-rank trend  $\chi^2(1) = 60.56$ ,  $p < 0.001$ ).

## Discussion

This study used data from the Framingham Heart Study to evaluate the associations of diabetes and total cholesterol with risk of MI/CHD among 4,354 participants.

Consistent with prior literature, diabetes was strongly associated with increased MI/CHD hazard. In univariate analyses, participants with diabetes had more than a fourfold higher hazard compared to non-diabetic participants (HR = 4.14, 95% CI: 3.04–5.63,  $p < 0.001$ ). Univariate analysis showed that a 10 mg/dL increase in total cholesterol levels was associated with 7% increase in the hazard of MI/CHD (HR = 1.07; 95% CI: 1.06–1.09;  $p < 0.001$ ).

In the final multivariable model, diabetes remained independently associated with MI/CHD risk, although its effect was attenuated (HR = 3.0, 95% CI: 2.19–4.10,  $p < 0.001$ ). Linearity of total cholesterol was assessed using Martingale residuals, which indicated non-linearity on the original scale. A log transformation improved the functional form and was therefore used in subsequent models. Schoenfeld residuals suggested a violation of the proportional hazards assumption for cholesterol. Accordingly, a time-varying coefficient was incorporated to account

for this non-proportionality. The final Cox model included log-transformed cholesterol as both a main effect and a time-varying covariate. Total cholesterol was strongly associated with MI/CHD risk at baseline per log-unit increase (HR = 18.73, 95% CI: 6.37–55.12) with the cholesterol hazard ratio diminishing per year (tvc HR = 0.64, 95% CI: 0.41–0.99). For example, at 10 years, one log-unit increase in log-transformed cholesterol is associated with approximately  $18.73 \times 0.64^{\log(10)} = 6.7$  times the hazard of MI/CHD, suggesting that total cholesterol is particularly detrimental early in follow-up. Age was identified as a confounder for both modifiable exposures and thus retained in the final model. No transformation was required as the Martingale residual plot of age failed to provide evidence of non-linearity. Holding all other variables in the model constant, each one-year increase in age was associated with a 4.8% increase in the hazard of MI/CHD (HR = 1.048, 95% CI: 1.04–1.06,  $p < 0.001$ ). The model was stratified by sex due to violation of the proportional hazards assumption identified through Schoenfeld residual testing. While this approach allows the baseline hazard to vary across men and women without imposing proportionality, it prevents estimation of a hazard ratio for sex.

There was no evidence of effect modification by diabetes status on hazard of MI/CHD found ( $p_{\text{interaction}} = 0.136$ ). When cholesterol was categorized, a dose-response relationship emerged, with progressively higher hazards in the “at risk” and “dangerous” groups relative to the “healthy” group.

Several limitations should be noted. Firstly, the Framingham cohort was predominantly white and from a single city, limiting generalizability. Secondly, although a test for confounding was assessed, residual confounding from unmeasured factors such as diet or medication use may persist. Thirdly, risk factors were measured at baseline only. Additionally, the large hazard ratio for log-transformed cholesterol at baseline should be interpreted cautiously given the scale of transformation and wide confidence interval. Finally, as this was an observational study, causality cannot be definitively established.

## Contributions

Both members collaborated extensively on the completion of this project. Prior to Dr. Kawaguchi providing the cleaned dataset, Chiamaka independently requested access to the Framingham dataset through BioLINCC and performed the initial data cleaning. Michelle played a key role in organizing timelines and coordinating task delegation.

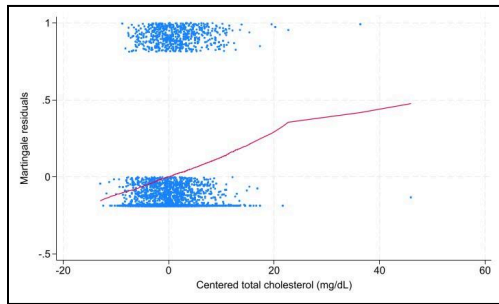
Responsibilities were divided such that Michelle primarily completed Problems 3a–3d, while Chiamaka was responsible for Problems 3f–3h. Both members provided support to one another when challenges arose; for example, Michelle assisted Chiamaka extensively with debugging code. All sections were jointly conceptualized, with Michelle taking the lead on editing the methods and results, and Chiamaka taking the lead on writing the discussion and contribution sections. Michelle was instrumental in the formatting of the final document and STATA codes.

Overall, both members worked collaboratively to ensure the timely and successful completion of the project.

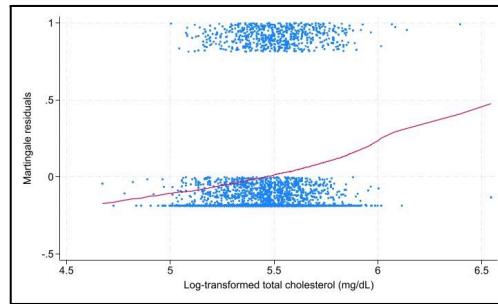
## Supplementary Materials

**Figure S1.** Martingale Residual Plots Assessing Linearity of Total Cholesterol in the Cox Proportional Hazards Model

**Panel A:**



**Panel B:**



LOWESS smoothers (pink line) assess linearity of total cholesterol in the Cox proportional hazards model. Panel A: centered total cholesterol; Panel B: log-transformed total cholesterol.

**Table S1.** Assessment of confounding for the association between diabetes, total cholesterol, and MI/CHD

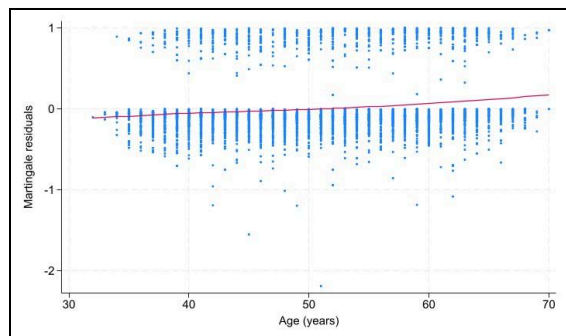
|                                     | HR (Diabetes) | % Change     | HR (Cholesterol) | % Change     |
|-------------------------------------|---------------|--------------|------------------|--------------|
| Base (diabetes + cholesterol, only) | 3.91          | —            | 7.02             | —            |
| + Age                               | 3.22          | <b>17.6%</b> | 4.36             | <b>37.9%</b> |
| + Sex                               | 3.82          | 2.8%         | 8.99             | <b>28.1%</b> |
| + Education                         | 3.73          | 4.6%         | 7.13             | 1.5%         |

Percent change calculated relative to the base model. Bolded values indicate a  $\geq 10\%$  change in the hazard ratio, providing evidence of confounding. HR = hazard ratio; MI/CHD = myocardial infarction or coronary heart disease.

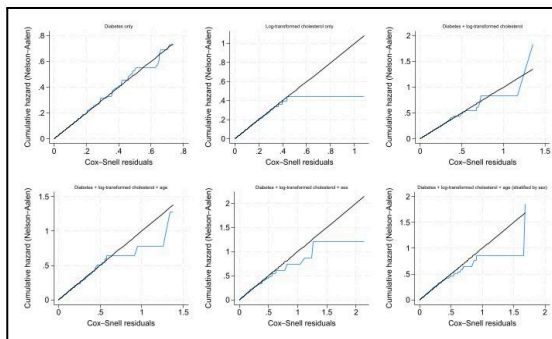


**Figure S2.** Model Diagnostics for the Stratified Cox Proportional Hazards Model of MI/CHD Risk

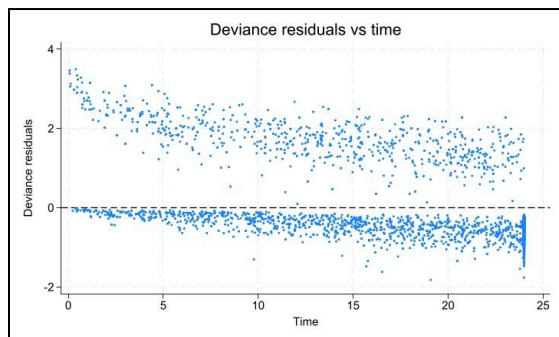
**Panel A:**



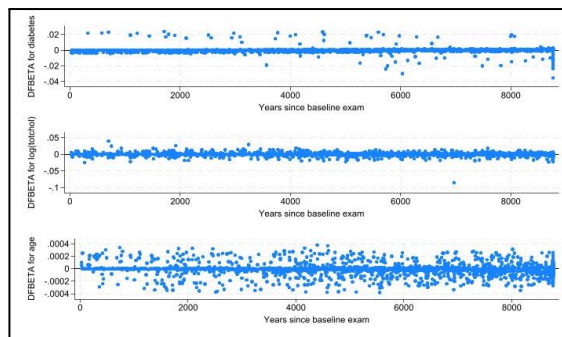
**Panel B:**



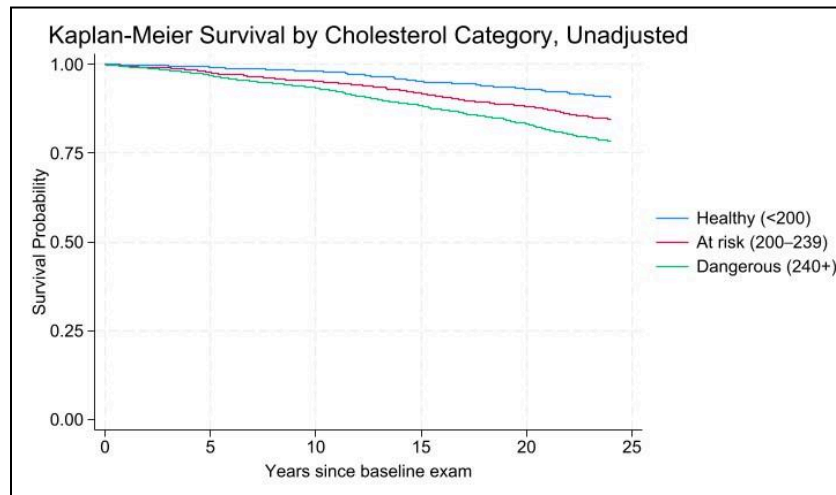
**Panel C:**



**Panel D:**



Martingale residual plots for age show no evidence of non-linearity (Panel A). Cox-Snell residual plots indicated that overall model fit improved with the addition of diabetes, log cholesterol, and age, stratified by sex (Panel B). Deviance residual plots showed that most observations were centered around zero (Panel C). DFBETA plots indicated that individual observations had minimal influence on model coefficients (Panel D).

**Figure S3.** Kaplan-Meier Survival Curve for Categorical Cholesterol

Kaplan-Meier survival curves for time to first myocardial infarction or coronary heart disease death, by total cholesterol category. Cholesterol categories defined as healthy (<200 mg/dL), at risk (200–239 mg/dL), and dangerous (≥240 mg/dL).

**Table S2.** Unadjusted and Adjusted Associations of Categorical Total Cholesterol with MI/CHD Risk

| Cholesterol Category    | Unadjusted HR      | p-value | Adjusted HR        | p-value |
|-------------------------|--------------------|---------|--------------------|---------|
| Healthy (<200 mg/dL)    | Reference          | —       | Reference          | —       |
| At risk (200–239 mg/dL) | 1.75 (1.33 – 2.30) | <0.001  | 1.53 (1.17 – 2.01) | 0.002   |
| Dangerous (≥240 mg/dL)  | 2.54 (1.96 – 3.28) | <0.001  | 2.22 (1.71 – 2.87) | <0.001  |

HR (CI); HR = hazard ratio; Adjusted model stratified by sex and includes diabetes status and age.